

3 Arteries, capillaries and veins are three types of blood vessel.

(a) Table 3.1 shows some features of these three types of blood vessel.

Complete Table 3.1 by using a tick (✓) if the feature is present in the blood vessel and a cross (X) if the feature is absent from the blood vessel.

Put a tick (✓) or a cross (X) in every box.

Table 3.1

feature	artery	capillary	vein
smooth muscle			
endothelium			
tunica media			

[3]

(b) Fig. 3.1 shows a transmission electron micrograph of part of an alveolus and part of the adjacent capillary.

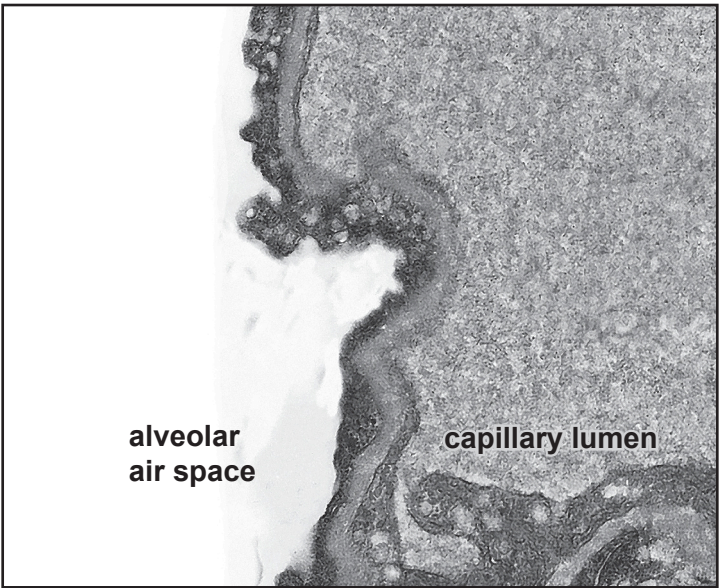


Fig. 3.1

(i) Draw **two** labelled arrows on Fig. 3.1 to show the direction of movement of oxygen and the direction of movement of carbon dioxide during gas exchange in the lungs. [1]

- (ii) Suggest **and** explain how a steep oxygen concentration gradient is maintained in the lungs.

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..... [4]

- (c) The passage outlines the roles of blood vessels associated with the heart.

Complete the passage by using the most appropriate scientific terms.

The carries blood to the left atrium. After passing from the left atrium to the left ventricle, blood is pumped into the aorta. The aorta is one of two large arteries that carry blood away from the ventricles of the heart. Blood that leaves the heart to enter these arteries must pass through the valves.

Oxygenated blood is supplied to the cardiac muscle cells through the arteries.

[3]

[Total: 11]